

Lecture 3/Module 1

DC Circuit Analysis and Theorems

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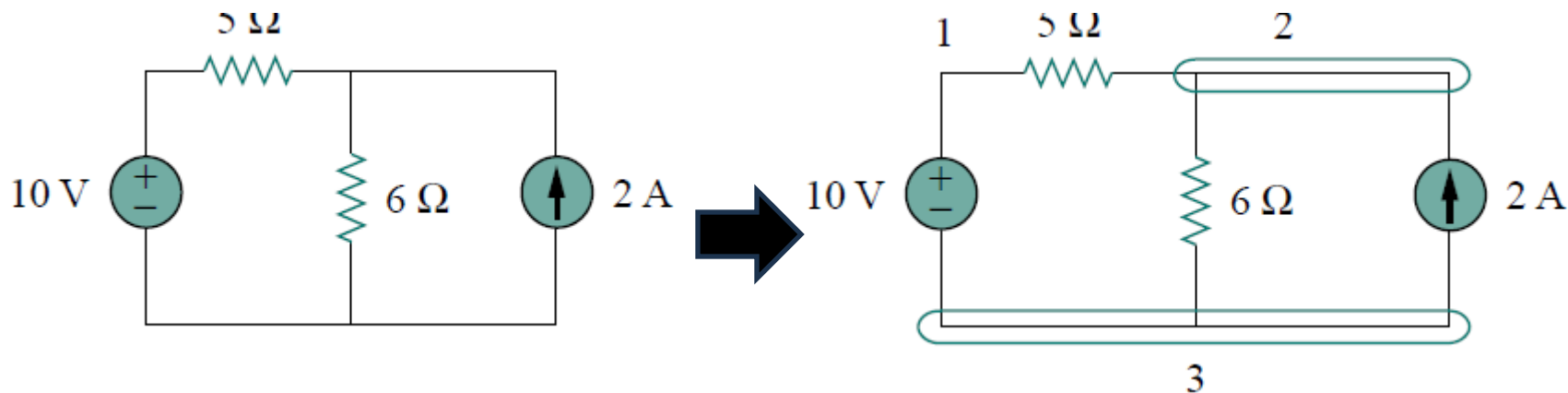
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Lecture Topics



- Degree of a Node, Reference, Non reference Node
- Kirchoff's Laws (KCL, KVL)

Identify the Nodes and Branches in the Circuit.



Branches: 4 nos.

i. 10 V voltage source,

ii. 5 Ω resistor

iii. 6 Ω resistor

iv. 2A current source

Nodes: 1, 2, 3 (Total 3 nos.)

Degree 2: Node 1 (Simple Node)

Degree 3: Node 2, 3

Non reference Node: Node 1, 2

Reference Node: Node 3

Non-Reference with Degree 3: Node 2 (Principal Node)

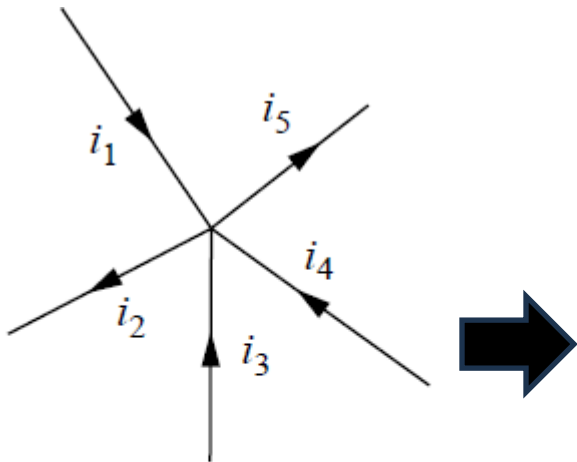
Kirchhoff's Current Law (KCL)

Kirchhoff's current law (KCL) states that the algebraic sum of currents entering a node (or a closed boundary) is zero.

Mathematically, KCL implies that

$$\sum_{n=1}^N i_n = 0$$

**Conservation of Charge OR
Current accumulation at any
node is zero**



Consider the node in Fig. 2.16. Applying KCL gives

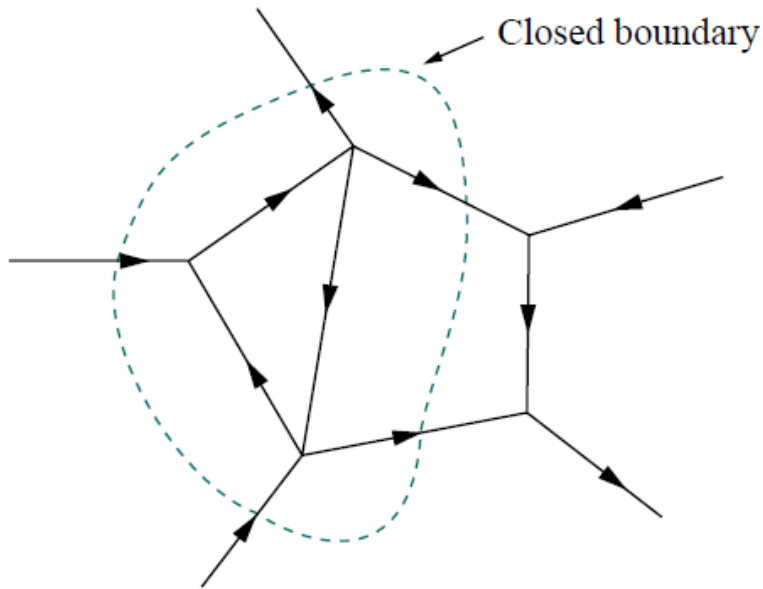
$$i_1 + (-i_2) + i_3 + i_4 + (-i_5) = 0$$

$$i_1 + i_3 + i_4 = i_2 + i_5$$

Figure 2.16 Currents at a node illustrating KCL.

The sum of the currents entering a node is equal to the sum of the currents leaving the node.

KCL on Closed Boundary



Note that KCL also applies to a closed boundary. This may be regarded as a generalized case, because a node may be regarded as a closed surface shrunk to a point. In two dimensions, a closed boundary is the same as a closed path. As typically illustrated in the circuit of Fig. 2.17, the total current entering the closed surface is equal to the total current leaving the surface.

Figure 2.17 Applying KCL to a closed boundary.

Application of KCL

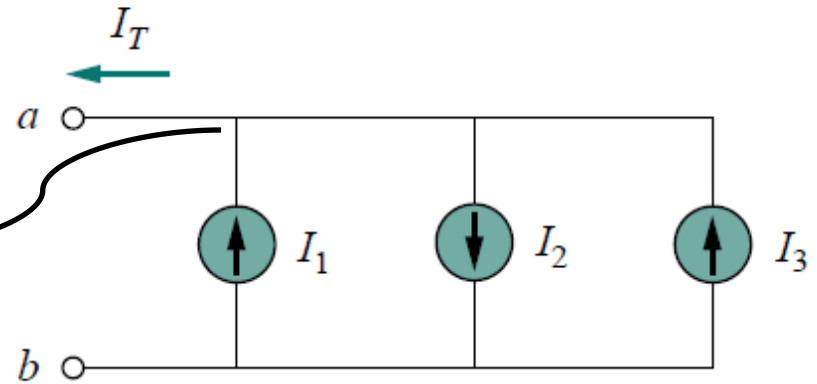
When current sources are connected in parallel

$$I_T + I_2 = I_1 + I_3$$

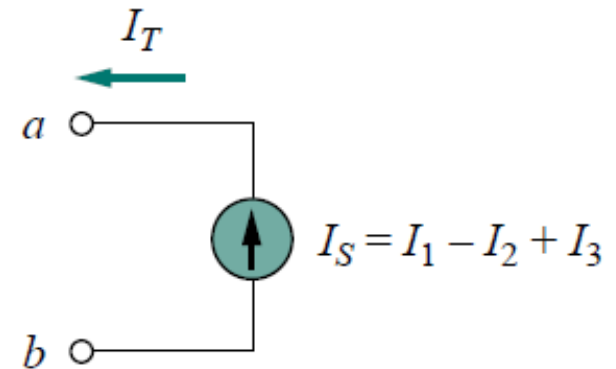


$$I_T = I_1 - I_2 + I_3$$

A circuit cannot contain two different currents, I_1 and I_2 , in series, unless $I_1 = I_2$; otherwise KCL will be violated.

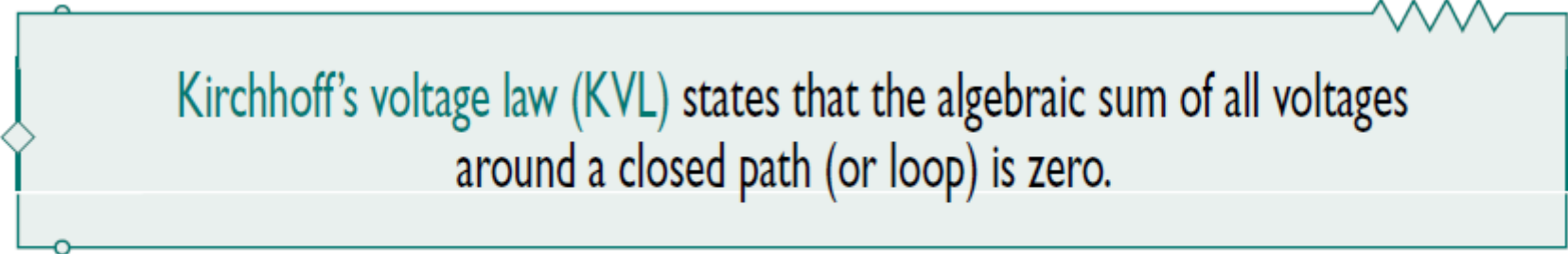


(a)



(b)

Kirchhoff's Voltage Law (KVL)



Kirchhoff's voltage law (KVL) states that the algebraic sum of all voltages around a closed path (or loop) is zero.

Expressed mathematically, KVL states that

$$\sum_{m=1}^M v_m = 0$$



Energy Conservation

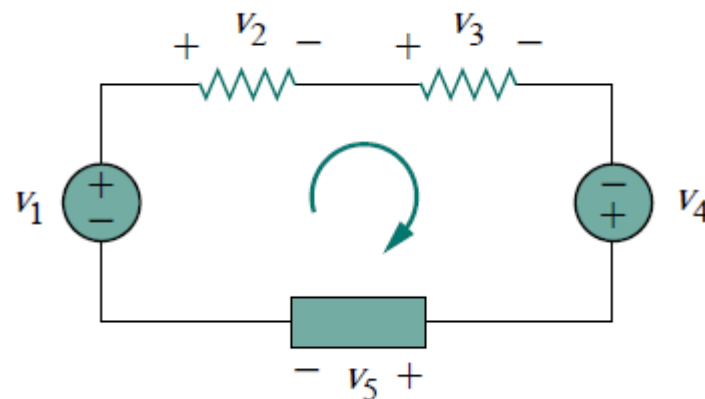
Sign Convention while finding KVL equation

Applying KVL

$$-v_1 + v_2 + v_3 - v_4 + v_5 = 0$$

OR

$$v_2 + v_3 + v_5 = v_1 + v_4$$



Sum of voltage drops = Sum of voltage rises

Voltage sign depends on the polarity of element encountered first while moving around the loop clockwise/counterclockwise direction

Application of KVL

When voltage sources are connected in parallel

$$-V_{ab} + V_1 + V_2 - V_3 = 0$$



$$V_{ab} = V_1 + V_2 - V_3$$

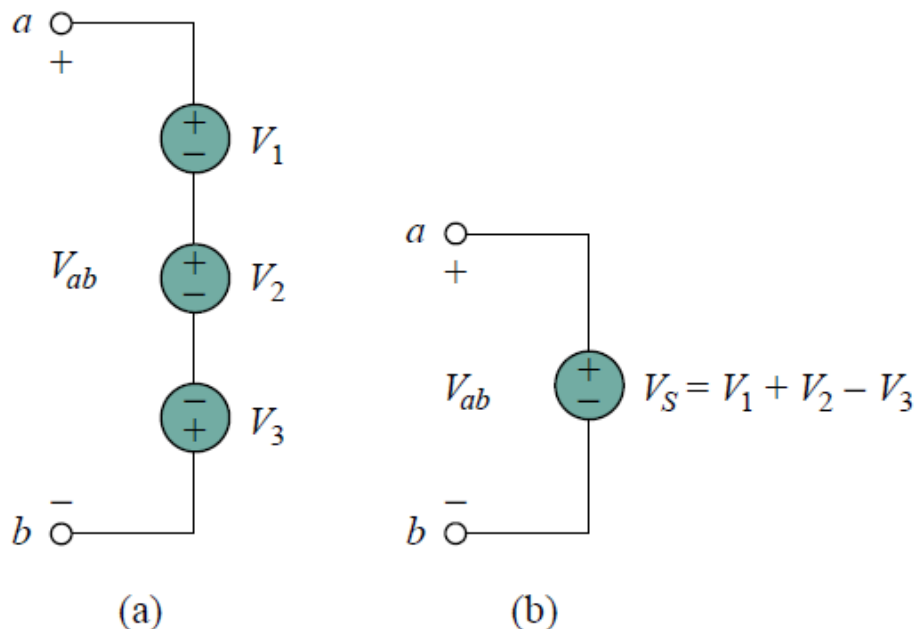


Figure 2.20

Voltage sources in series:
(a) original circuit, (b) equivalent circuit.

To avoid violating KVL, a circuit cannot contain two different voltages V_1 and V_2 in parallel unless $V_1 = V_2$.

Summary

KCL

- KCL is based on law of conservation of electronic charge
- Charge cannot accumulate at a node
- KCL holds for both planar and non-planar circuits

KVL

- KVL follows from the law of conservation of energy
- Total energy gained by charges equals total energy lost when they traverse a closed path
- KVL holds for both planar and non-planar circuits

Assignment

What is Planar Circuit and what is Non-Planar Circuit ?? Explain with diagram

Submit by Tuesday (04/08/2026)

Thank You